LESSON 8: Spectral Measures and the Spectral Theorem
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1. Spectral measures

In this lesson, we illustrate the version of the spectral theorem that is used to state the axioms of quantum mechanics. This demonstration also showcases the full power of the spectral theorem and its various applications, both within and outside of quantum mechanics. We now recall the Spectral Theorem for Unbounded operators as functional calculus (version 1).

Theorem 1.1 (Spectral Theorem for Unbounded operators as functional calculus (version 1))**.** *Let $A : D(A) \subset H \to H$ be an unbounded self-adjoint operator, where H is a complex separable Hilbert space. Then there exists a unique function $\psi : B(\mathbb{R}) \to \mathcal{B}(H)$ that has the following properties:*

(I) $\|\psi(g)\|_{\mathcal{B}(H)} \leq \|g\|_\infty, \quad \forall g \in B(\mathbb{R})$.

(II) *ψ is a $*$-homomorphism.*

(III) *If g_n and $g \in B(\mathbb{R})$ with $g_n \to g$ for $t \in \mathbb{R}$ and $\sup_{n\in\mathbb{N}} |g_n| < \infty$, then*

$$\psi(g_n)x \to \psi(g)x, \quad \forall x \in H.$$

We now present another version of the spectral theorem that will be useful for applications. The introduction of spectral measures allows us to clearly express the postulate of wave function collapse and to represent the solutions of the Schrödinger equation through projectors.

Definition 1.2 (Definition of spectral measures)**.** *A spectral measure is a family $E(t)$, $t \in \mathbb{R}$ of bounded linear operators in $\mathcal{B}(H)$ that satisfies the conditions*

(I) *Is a self-adjoint projector, i.e.,*

$$E(t) = E^2(t) = E^*(t), \quad \forall t \in \mathbb{R}$$

(II)

$$E(t)E(s) = E(\min(\{t, s\}), \quad \forall t, s \in \mathbb{R}$$

(III)

$$\lim_{t\to+\infty} E(t)x = x, \quad \lim_{t\to-\infty} E(t)x = 0, \quad \lim_{t\to s^+} E(t)x = E(s)x.$$

Theorem 1.3 (Properties of the spectral measures)**.** *The spectral measures has the following properties:*

(A)

$$E(t)E(s) = E(s)E(t) \quad \forall t, s \in \mathbb{R}$$

(B) *The function*

$$t \mapsto (E(t)x, x)$$

is nonnegative, increasing and continue on the right (i.e., is a function of bounded variation).

Remark 1.4. *Thanks to property (B) it is possible to define the integral with respect to the spectral measures.*

Remark 1.5 (The generalized Stieltjes integral)**.** *We observe that if we take $f \in B(\mathbb{R})$ bounded and measurable on $\mathbb{R}$ in order to define the integral*

$$\int_{\mathbb{R}} f(t)d(E(t)x,y), \quad x,y \in H$$

we have to prove that $t \mapsto (E(t)x,y)$ is a function with bounded variation. This is true since $(E(t)x,y)$ is the finite some of functions

$$(E(t)z,z) \text{ where z take the values } x \pm y \text{ and } x \pm iy$$

that follows from the relation

$$(E(t)x,y) = \frac{1}{4}\Big((E(t)(x+y),x+y) - (E(t)(x-y),x-y)\Big)$$
$$-\frac{i}{4}\Big((E(t)(x+iy),x+iy) - (E(t)(x-iy),x-iy)\Big)$$

is called polarization formula.

Definition 1.6. *Let $f \in B(\mathbb{R})$ (bounded and measurable on $\mathbb{R}$) we set*

$$\int_{\mathbb{R}} f(t)d(E(t)x,y) := \lim_{p\to\infty}\int_{-p}^{p} f(t)d(E(t)x,y), \quad x,y \in H \quad p>0$$

Remark 1.7. *Let $f \in B(\mathbb{R})$ (bounded and measurable). One can prove that the limit exists and also that there exists a constant $C_E > 0$ such that*

$$\Big|\int_{\mathbb{R}} f(t)d(E(t)x,y)\Big| \le C_E\|x\|\|y\|, \quad x,y \in H.$$

We arrive to the definition of $\int_{\mathbb{R}} f(t)dE(t)$ by the following theorem.

Theorem 1.8. *Let $E(t)$, $t \in \mathbb{R}$, be a spectral measure. Then, for every $f \in B(\mathbb{R})$, there exists a unique bounded linear operator $\Psi(f)$ such that*

$$(\Psi(f)x,y) = \int_{\mathbb{R}} f(t)d(E(t)x,y), \quad x,y \in H$$

We indicate the linear operator $\Psi(f)$ as

$$\Psi(f) = \int_{\mathbb{R}} f(t)dE(t).$$

Proof. We observe that

$$B_f(x,y) := \int_{\mathbb{R}} f(t)d(E(t)x,y), \quad x,y \in H$$

for every $f \in B(\mathbb{R})$ fixed, is a sesquilinear form (or hermitian). Since it is bounded

$$\Big|\int_{\mathbb{R}} f(t)d(E(t)x,y)\Big| \le C_E\|x\|\|y\|, \quad x,y \in H$$

there exists, thanks to Riesz representation theorem, a unique bounded linear operator $\Psi(f)$ on H such that

$$(\Psi(f)x, y) = B_f(x, y), \ x, y \in H.$$

□

Remark 1.9. *The integral*

$$\int_{\mathbb{R}} f(t)dE(t)$$

is an operator that is, by definition, identified by the infinite numbers

$$\int_{\mathbb{R}} f(t)d\big(E(t)x, y\big) = \big(\int_{\mathbb{R}} f(t)dE(t)x, y\big)$$

where $x, y \in H$ vary in all possible ways.

Remark 1.10 (The problem of the domain of an unbounded self-adjoint operators for functions that are not bounded)**.** *Observe that for unbounded self-adjoint operator*

$$A : D(A) \subset X \to X$$

since function $f \in B(\mathbb{R})$ are bounded the operator

$$\int_{\mathbb{R}} f(t)dE(t)$$

is bounded.

The definition can be extended to unbounded functions, in this case we cannot consider

$$\int_{\mathbb{R}} f(t)d(E(t)x, y) \quad \text{for all} \quad x, y \in H.$$

We have to introduce a domain for the operators. Precisely, in this case the operator turns out to be unbounded and the natural domain is

$$D\Big(\int_{\mathbb{R}} f(t)dE(t)\Big) := \big\{x \in H \ : \ \int_{\mathbb{R}} |f(t)|^2 d(E(t)x, x) < \infty\big\}$$

for every $x \in D\Big(\int_{\mathbb{R}} f(t)dE(t)\Big)$ the vector $\int_{\mathbb{R}} f(t)dE(t)x$ is uniquely identified by

$$(\int_{\mathbb{R}} f(t)dE(t)x, y) = \int_{\mathbb{R}} f(t)d(E(t)x, y), \quad y \in H.$$

Theorem 1.11 (Spectral measure of a self-adjoint operator)**.** *Let $A : D(A) \subset H \to H$ be a self adjoint operator and let ψ*

$$\psi : B(\mathbb{R}) \to \mathcal{B}(H).$$

be as in Theorem 1.1 (Spectral Theorem for Unbounded operators as functional calculus (version 1)). Then, we define:

$$F(t) := \psi(\chi_{(-\infty,t]})$$

where

$$\chi_{(-\infty,t]}(r) = \begin{cases} 1 & \text{if } \ r \leq t, \\ 0 & \text{if } \ r > t. \end{cases}$$

The operator $F(t)$ is the spectral measure of A. That is $F(t)$ satisfies the properties:

(I) *Is a self-adjoint projector, i.e.,*

$$F(t) = F^2(t) = F^*(t), \quad \forall t \in \mathbb{R}$$

(II)

$$F(t)F(s) = F(\min(\{t,s\}), \quad \forall t,s \in \mathbb{R}$$

(III)

$$\lim_{t\to+\infty} F(t)x = x, \quad \lim_{t\to-\infty} F(t)x = 0, \quad \lim_{t\to s^+} F(t)x = F(s)x.$$

Proof. Proof of Property (I): $F(t) = F^2(t) = F^*(t), \quad \forall t \in \mathbb{R}$.

From the definition of $\chi_{(-\infty,t]}(r)$ it follows that

$$(\chi_{(-\infty,t]}(r))^2 = \chi_{(-\infty,t]}(r) = \overline{\chi_{(-\infty,t]}(r)}$$

Since ψ is a $*$-homomorphism, i.e.,

- $\psi(\lambda g_1 + \mu g_2) = \lambda\psi(g_1) + \mu\psi(g_2)$
- $\psi(g_1 g_2) = \psi(g_1)\psi(g_2)$
- $\psi(1) = I$
- $\psi(\overline{g_1}) = \psi(g_1)^*$

so from the definition of

$$F(t) := \psi(\chi_{(-\infty,t]})$$

we have that

$$F^2(t) = F(t) = F(t)^*, \quad t \in \mathbb{R}$$

So F is an self-adjoint projector.

We prove property (II): $F(t)F(s) = F(\min(\{t,s\}), \quad \forall t,s \in \mathbb{R}$.

Observe that for $t,s \in \mathbb{R}$ we have the relation

$$\chi_{(-\infty,t]}(r) \ \cdot \ \chi_{(-\infty,s]}(r) = \chi_{(-\infty,\min\{t,s\}]}(r), \quad \in \mathbb{R}$$

by the property of ψ

$$\psi\big(\chi_{(-\infty,t]}(r) \ \cdot \ \chi_{(-\infty,s]}(r)\big) = \psi(\chi_{(-\infty,\min\{t,s\}]}(r))$$

and so

$$\psi\big(\chi_{(-\infty,t]}(r)\big) \ \cdot \ \psi\big(\chi_{(-\infty,s]}(r)\big) = \psi(\chi_{(-\infty,\min\{t,s\}]}(r))$$

which means

$$F(t)F(s) = F(\min(\{t,s\}), \quad \forall t,s \in \mathbb{R}.$$

We prove property (III) $\lim_{t\to+\infty} F(t)x = x$, $\lim_{t\to-\infty} F(t)x = 0$, $\lim_{t\to s^+} F(t)x = F(s)x$.

We verify the for all $x \in H$

$$\lim_{t\to s^+} F(t)x = F(s)x.$$

Let t_n be a decreasing sequence such that $t_n \to s$. Then we have

$$\lim_{n\to\infty} \chi_{(-\infty,t_n]}(r) = \chi_{(-\infty,s]}(r) \quad \text{and} \quad \sup_{n\in\mathbb{N}} \|\chi_{(-\infty,t_n]}\|_\infty < \infty.$$

From condition [(III)] of the spectral theorem in functional calculus form, that is Theorem 1.1),

That is: If g_n and $g \in B(\mathbb{R})$ with $g_n \to g$ for $t \in \mathbb{R}$ and $\sup_{n\in\mathbb{N}} |g_n| < \infty$, then

$$\psi(g_n)x \to \psi(g)x, \quad \forall x \in H.$$

we have

$$\lim_{n\to\infty} \psi(\chi_{(-\infty,t_n]})x = \psi(\chi_{(-\infty,s]})x$$

and so

$$\lim_{t\to s^+} F(t)x = F(s)x.$$

The other conditions $\lim_{t\to+\infty} F(t)x = x$ and $\lim_{t\to-\infty} E(t)x = 0$ follows similarly.

□

Theorem 1.12 (The Spectral Theorem second form (via spectral measures))**.** *Let*

$$A : D(A) \subset H \to H$$

be a self-adjoint operator on a separable Hilbert space H.
Let

$$E(t) := \psi(\chi_{(-\infty,t]}), \quad t \in \mathbb{R}$$

be its spectral measure of A defined via the spectral theorem in form 1 (functional calculus). Then, for every $f \in B(\mathbb{R})$ we have

$$(\Psi(f)x, y) = \int_{\mathbb{R}} f(t)d(E(t)x, y), \quad x, y \in H$$

or equivalently

$$\psi(f) = \int_{\mathbb{R}} f(t)dE(t).$$

Moreover, we have the equality

$$\|\psi(f)x\|^2 = \int_{\mathbb{R}} |f(t)|^2 d(E(t)x, x) \quad \text{forall } x \in H \text{ and } \ f \in B(\mathbb{R}).$$

Proof. It follows as a consequence the spectral theorem in the functional calculus form. We only consider the case of bounded linear operators we omit the cased of unbounded operators.

Let $A : H \to H$ be self-adjoint. We take an interval $[a, b]$ that contains $\sigma(A)$.

For every x, $y \in H$ we have defined the functions of bounded variation $F_{x,y}(t)$ on $[a, b]$ and we defined the operator ψ by the equality

$$(\psi(f)x, y) = \int_a^b f(t)dF_{x,y}(t), \quad f \in B(\mathbb{R}), \quad x, y \in H.$$

- If we extend the definition of $F_{x,y}(t)$ on $[a,b]$ to $\mathbb{R}$ as follows
$$F_{x,y}(t) := F_{x,y}(b), \quad for \quad t > b \quad and \quad F_{x,y}(t) := F_{x,y}(a), \quad for \quad t < a$$
it is clear that
$$\int_{\mathbb{R}} f(t) dF_{x,y}(t) = \int_a^b f(t) dF_{x,y}(t), \quad f \in B(\mathbb{R}), \quad x, y \in H$$
- so we get
$$(\psi(f)x, y) = \int_{\mathbb{R}} f(t) dF_{x,y}(t), \quad f \in B(\mathbb{R}), \quad x, y \in H.$$
- now fix $s \in \mathbb{R}$ and take the characteristic function $f(t) = \chi_{(-\infty,s]}(t)$,
- by the definition of $E(t) := \psi(\chi_{(-\infty,s]}(t))$ we have
$$(\psi(\chi_{(-\infty,s]}(t))x, y) = (E(t)x, y)$$
- and so, by the properties of the Stieltjes integral
$$\int_{\mathbb{R}} \chi_{(-\infty,s]}(t) dF_{x,y}(t) = F_{x,y}(s).$$
This shows that
$$(E(s)x, y) = F_{x,y}(s)$$
so replacing we get
$$(\psi(f)x, y) = \int_{\mathbb{R}} f(t) dF_{x,y}(t) = \int_{\mathbb{R}} f(t) d(E(t)x, y), \quad f \in B(\mathbb{R}), \quad x, y \in H,$$
which is the equality we required.

□

We can now state without proofs the most general version of he spectral theorem for bounded or unbounded self-adjoint operator.

Theorem 1.13 (The Spectral theorem (general version) for unbounded functions and with spectral measures). *Let A be a self-adjoint operator on a Hilbert space H. Then there exists a unique spectral measure E_A on $\mathbb{R}$ (on the Borel sets of $\mathbb{R}$) such that*
$$A = \int_{\mathbb{R}} \lambda dE_A(\lambda),$$
with domain
$$D(A) = \{x \in H \ : \ \int_{\mathbb{R}} |\lambda|^2 d(E(\lambda)x, x) < \infty \ \}.$$
So we can use the spectral theorem as a functional calculus for functions
$$f : \mathbb{R} \to \mathbb{C} \cup \{\infty\}$$
that are E_A-a.e finite Borel functions we write
$$f(A) = \int_{\mathbb{R}} f(\lambda) dE(\lambda)$$

with domain

$$D(f(A)) = \{x \in H \ : \ \int_{\mathbb{R}} |f(\lambda)|^2 d(E(\lambda)x, x) < \infty \ \}.$$

2. Groups of unitary operators: Stone's theorem

Definition 2.1 (Group of unitary operators)**.** *A group of unitary operators is a family of unitary operators $U(t)$ depending on a real parameter $t \in \mathbb{R}$ that satisfies the conditions*

- $U(t+s) = U(t)U(s), \quad t, s \in \mathbb{R}$
- $U(0) = I$.

We say that the group is strongly continuous if for all $x \in H$ and

$$\lim_{s \to 0} U(t+s)x = U(t)x, \quad \forall x \in H, \quad \forall t \in \mathbb{R}.$$

Remark 2.2 (Group of unitary operators: consequences)**.** *Observe that*

$$U(t)U(s) = U(t+s) = U(s+t) = U(s)U(t)$$

and since $U(t)$ is unitary $UU^* = U^*U = I$

$$U(t)U(-t) = U(t)U(-t) = I \quad \Rightarrow \quad U(t)^{-1} = U(t)^* = U(-t)$$

Definition 2.3 (e^{itA} defined via the spectral theorem)**.** *Recall that by the spectral theorem if A is a self-adjoint operator with spectral measures then the operator e^{itA} is defined as*

$$e^{itA} := \int_{\mathbb{R}} e^{it\lambda} dE(\lambda).$$

Theorem 2.4. *Let A be a self-adjoint operator on a Hilbert space H. Then $U := e^{itA}$, $t \in \mathbb{R}$ is a strongly continuous one-parameter unitary group on H. The operator A is uniquely determined by $U(t)$, that is the domain of A is given by*

$$D(A) = \{x \in H \quad : \quad \lim_{h \to 0} h^{-1}(U(h) - I) \ \ exists\}.$$

and

$$iAx = \lim_{h \to 0} h^{-1}(U(h) - I) \ \ \forall x \in D(A).$$

Moreover, it is

$$\frac{d}{dt} U(t)x = iAU(t)x, \quad \forall x \in D(A).$$

Theorem 2.5 (Stone's theorem)**.** *If U is a strongly continuous one-parameter unitary group on H then there exists a unique self-adjoint operator on A such that $U(t) = e^{itA}$ for all $t \in \mathbb{R}$.*

Definition 2.6. *The operator iA is called the infinitesimal generator of the unitary group U.*

3. Applications of the spectral theorem

Example 3.1 (Diagonal operator)**.** *Consider the diagonal operator*

$$A : D(A) \subset L^2(a,b) \to L^2(a,b)$$

with eigenvalues $\lambda_i \in \mathbb{R}$ *and associated eigenfunctions* $e_i(x)$, $i \in \mathbb{N}$ *that are an orthonormal basis of* $L^2(a,b)$

$$A\psi(x) = \sum_{i=1}^{\infty} \lambda_i (\psi, e_i) e_i(x),$$

with domain

$$D(A) = \{\psi \in L^2(a,b) \quad : \quad \sum_{i=1}^{\infty} \lambda_i^2 |(\psi, e_i)|^2 < \infty\}.$$

Observe that for $f(\lambda) = \lambda$ *since there are only the eigenvalues* λ_i *it is*

$$\psi(f) = A = \int_{\mathbb{R}} \lambda dE(\lambda) = \sum_{i=1}^{\infty} \lambda_i (\cdot, e_i) e_i.$$

If we consider $\lambda_1 < \lambda_2 < ..... < \lambda_n < \lambda_{N+1} < ....$ *these are self-adjoint projectors:*

$$E(\lambda)\psi = \sum_{n:\lambda_n \leq \lambda} \lambda_i (\psi, e_i) e_i.$$

Via the spectral theorem it is

$$e^{itA}\psi(x) = \sum_{i=1}^{\infty} e^{it\lambda_i} (\psi, e_i) e_i(x), \quad \psi \in L^2(a,b).$$

For the sake of simplicity consider operators with $\lambda_i > 0$. *For* $\alpha > 0$ *we can also define the fractional powers* A^α *with the domain*

$$D(A^\alpha) = \{\psi \in L^2(a,b) \quad : \quad \sum_{i=1}^{\infty} \lambda_i^{2\alpha} |(\psi, e_i)|^2 < \infty\}$$

$$A^\alpha \psi(x) = \sum_{i=1}^{\infty} \lambda_i^\alpha (\psi, e_i) e_i(x), \quad \psi \in D(A^\alpha).$$

Recall that Schrödinger equation written more concisely in operator form, associated with the Hilbert space $\mathcal{H}$:

$$i\hbar \frac{d\Psi(t)}{dt} = H\Psi(t), \qquad H : D(H) \subset \mathcal{H} \to \mathcal{H}$$

where $H = -\frac{\hbar^2}{2m}\Delta + V$ is the Hamiltonian operator.

Example 3.2 (Harmonic oscillator)**.** *For the quantum harmonic oscillator we have*

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2$$
$$H\psi_n(x) = E_n \psi_n(x)$$

The energy eigenvalues for the quantum harmonic oscillator are quantized:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, 3, \ldots$$

The quantum number n specifies the energy level, with $n = 0$ representing the ground state. The mathematical expression for these eigenfunctions is:

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \frac{1}{\sqrt{2^n n!}} \mathbf{H}_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right) e^{-\frac{m\omega x^2}{2\hbar}}.$$

The term $\mathbf{H}_n\left(\sqrt{\frac{m\omega}{\hbar}}x\right)$ involves the Hermite polynomials, $\mathbf{H}_n(x)$:

$$\mathbf{H}_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}(e^{-x^2}).$$

The spectral representation of the operator H in terms of the eigenfunction that are an orthonormal base in $L^2(\mathbb{R})$ is then given by

$$Hf(x) := \sum_{n=0}^{\infty} E_n (f, \psi_n)\psi_n(x)$$

with domain

$$D(H) = \{f \in L^2(\mathbb{R}) \quad : \quad \sum_{n=0}^{\infty} \left(n + \frac{1}{2}\right)^2 |(f, \psi_n)|^2 < \infty\}$$

this operator is self-adjoint.

Example 3.3. *Now consider*

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2 x^2 : D(H) \to L^2(\mathbb{R})$$

and the time dependent Schrödinger equation

$$i\hbar\frac{d}{dt}\psi(t) = H\psi(t),$$

$$\psi(0) = \varphi$$

where $\varphi \in \ L^2(\mathbb{R})$. We have

$$\psi(t, x) = e^{-i\frac{t}{\hbar}H}\varphi(x) = \sum_{n=0}^{\infty} e^{-i\frac{t}{\hbar}\left(n+\frac{1}{2}\right)\hbar\omega}(\varphi, \psi_n)\psi_n(x)$$

that satisfies Schrödinger equation with and it preserve the probability

$$\|\psi(t, \cdot)\|_{L^2(\mathbb{R})} = \|e^{-i\frac{t}{\hbar}H}\varphi(\cdot)\|_{L^2(\mathbb{R})} = \|\varphi\|_{L^2(\mathbb{R})}$$

$$e^{-i\frac{t}{\hbar}H} \quad \text{is a unitaty operator.}$$

Example 3.4 (The free particle and the continuous spectrum $\sigma_c(H)$)**.** *This is the case when we have only the continuous spectrum. In fact*

$$i\hbar\partial_t\psi(t,x) = -\frac{\hbar^2}{2m}\partial_x^2\psi(t,x), \qquad \psi(0,x) = \varphi(x), \qquad \varphi \in L^2(\mathbb{R}).$$

The eigenfunctions and the spectrum are described by the differential equation

$$-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x) = E\psi(x), \tag{3.1}$$

which can be rewritten as

$$\psi''(x) + \frac{2mE}{\hbar^2}\psi(x) = 0, \tag{3.2}$$

where the term $\frac{2mE}{\hbar^2}$ determines the nature of the solutions:

$$\frac{2mE}{\hbar^2} > 0 \quad or \quad \frac{2mE}{\hbar^2} < 0.$$

Considering the general solutions:

- *For $\frac{2mE}{\hbar^2} > 0$, set $k = \sqrt{\frac{2mE}{\hbar^2}}$. The solutions are:*

  $$\psi(x) = C_+e^{ikx} + C_-e^{-ikx}.$$

  However, for these functions, $\psi(x) \notin L^2(\mathbb{R})$ for any choice of constants $C_\pm$, meaning they are not square-integrable over the entire real line.
- *For $\frac{2mE}{\hbar^2} < 0$, define $k = i\kappa$ with $\kappa = \sqrt{\frac{2m|E|}{\hbar^2}}$. The solutions become:*

  $$\psi(x) = D_+e^{\kappa x} + D_-e^{-\kappa x}.$$

  Again, these functions are not square-integrable over $\mathbb{R}$.

The only solution that does not diverge at infinity is for $E > 0$. In this case we have the "improper eigenfunctions" the energy and the momentum are related by

$$E = \frac{p^2}{2m}$$

from which

$$p = \pm\sqrt{2mE}$$

so if we consider

$$\psi_p(x) = \frac{1}{\sqrt{2\pi\hbar}}e^{\frac{i}{\hbar}px}, \quad -\infty < p < +\infty$$

these are orthogonal "improper eigenfunctions" in the sense of distribution. With an abuse of notation we have

$$< \psi_p, \psi_{p'} >= \int_{\mathbb{R}} \psi_p(x)\overline{\psi_{p'}(x)}dx = \frac{1}{2\pi\hbar}\int_{\mathbb{R}} e^{\frac{i}{\hbar}px}e^{\frac{-i}{\hbar}p'x}dx = \frac{1}{2\pi\hbar}\int_{\mathbb{R}} 1 \cdot e^{\frac{i}{\hbar}(p-p')x}dx = \delta(p-p')$$

via the Fourier transform the solution is given by

$$\psi(t,x) = \frac{1}{\sqrt{2\pi\hbar}}\int_{\mathbb{R}} C(p)e^{\frac{i}{\hbar}\left(px - \frac{p^2}{2m}t\right)}dp$$

when we impose the initial condition

$$C(p) =< \psi_p, \varphi >= \int_{\mathbb{R}} \psi_p(x)\varphi(x)dx.$$

Also here we have

$$\|\psi(t,\cdot)\|_{L^2(\mathbb{R})} = \|\varphi\|_{L^2(\mathbb{R})}.$$

Example 3.5 (The hydrogen atom)**.** *One can also formalize the time-independent Schrödinger equation for the hydrogen atom is given by:*

$$H\Psi(r,\theta,\phi) = E\Psi(r,\theta,\phi)$$

where the Hamiltonian operator is:

$$H = -\frac{\hbar^2}{2m}\Delta + \frac{e^2}{4\pi\epsilon_0 r}$$

where r, θ, ϕ are the spherical coordinates, E is the energy eigenvalue and $\Psi(r,\theta,\phi)$ is the time independent wavefunction. To summarize the Eigenfunctions are

$$\Psi_{n,\ell,m}(r,\theta,\phi) = R_{n,\ell}(r)Y_\ell^m(\theta,\phi)$$

replacing $R_{n,\ell}(r)$ the radial part, and $Y_\ell^m(\theta,\phi)$ the spherical harmonics we have

$$\begin{aligned}\Psi_{n,\ell,m}(r,\theta,\phi) &= R_{n,\ell}(r)Y_\ell^m(\theta,\phi)\\ &= \left(\frac{2r}{na_0}\right)^n e^{-r/(na_0)} L_{n-\ell-1}^{(2\ell+1)}(2r/(na_0))(-1)^m \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}} P_\ell^m(\cos\theta)e^{im\phi}\end{aligned}$$

with

$$E_n = -\frac{m}{2n^2}\left(e^2/(4\pi\varepsilon_0\hbar)\right)^2.$$

They are such that

$$\int_{\mathbb{R}^3} |\Psi_{n,\ell,m}(t,r)|^2 = 1.$$

END OF THE DISSERTATION ON THE SPECTRAL THEOREM

CONCLUDING REMARKS AND SOME RESEARCH DIRECTIONS

(1) The axiomatic formulation of quantum mechanics is based on self-adjoint (Hermitian) operators in a Hilbert space (you will see more in the next courses in quantum mechanics).
(2) There is the possibility to merge special relativity and quantum mechanics; an example of this unification is the Dirac equation.
(3) A new spectral theory for vector operators has been developed based on a novel concept of spectrum that is suitable for these operators: the S-spectrum (active research here at the Politecnico di Milano with international research groups).

4. Axioms of quantum mechanics you will see in the course of quantum mechanics

In general, when we have a time-independent potential in the Hamiltonian

$$H = -\frac{\hbar^2}{2m}\Delta + V(x), \qquad D(H) \subset L^2(\mathbb{R}^3) \to L^2(\mathbb{R}^3) \tag{4.1}$$

the solution of the Schrödinger equation takes the form

$$\psi(t,x) = \sum_n C_n u_n(x) e^{-\frac{i}{\hbar}E_n t} + \int_{\sigma_c(H)} C(E) u_E(x) e^{-\frac{i}{\hbar}Et} dE \tag{4.2}$$

where the series is related to

$\sigma_p(H)$ the point spectrum: the eigenvalues E_n and eigenfunctions $u_n(x)$ that are in L^2 (these are the bound states of the system),

$\sigma_c(H)$ the part with an integral over the continuous spectrum E that represents the states where the particle is free (generalized eigenfunctions $u_E(x)$; here we work with distributions).

Via the spectral theorem in Hilbert space we can formulate quantum mechanics in an axiomatic way.

Original books by Dirac and John von Neumann.

- Dirac, Paul (1930), The Principles of Quantum Mechanics. The first and second editions of the book were published in 1930 and 1935.
- Mathematical Foundations of Quantum Mechanics: New Edition (Princeton Landmarks in Mathematics and Physics) New Edition by John von Neumann (Author), Nicholas A. Wheeler (Editor), Robert T. Beyer (Translator)

5. Special relativity and quantum mechanics

Operator theory enters many fields of science, and the way we form functions of operators is a crucial concept. We realize this from the conservation of

$$E^2 = \frac{P^2}{2m} + V,$$

replacing the the **energy operator** (Hamiltonian) is given by:

$$\hat{E} = i\hbar\frac{\partial}{\partial t}$$

The **momentum operator** in the position representation is:

$$\hat{\mathbf{p}} = -i\hbar\nabla$$

which leads to the Schrödinger equation. Similarly, we can extend this approach to the relativistic version of the Schrödinger equation, known as the Dirac equation. In fact,in relativistic mechanics, the total energy E of a particle is related to its momentum $\mathbf{p}$ and rest mass m by the equation:

$$E^2 = |\mathbf{p}|^2c^2 + m^2c^4$$

where:

- E is the total energy,
- $\mathbf{p}$ is the momentum,
- m is the rest mass of the particle,
- c is the speed of light.

We cannot directly replace the energy and momentum operators and interpret them in terms of probability density; instead, we must follow Dirac's procedure of linearizing the square root

$$E = \sqrt{|\mathbf{p}|^2c^2 + m^2c^4}$$

where $\mathbf{p} = (p_x, p_y, p_z)$ that is

$$E^2 = p_x^2c^2 + p_y^2c^2 + p_z^2c^2 + m^2c^4.$$

Dirac proposed to express the energy linearly in terms of momentum:

$$E = \alpha_1 p_x + \alpha_2 p_y + \alpha_3 p_z + \beta mc^2 = \boldsymbol{\alpha}\cdot\mathbf{p}c + \beta mc^2$$

where $\boldsymbol{\alpha} = (\alpha_x, \alpha_y, \alpha_z)$ and β cannot be complex numbers but matrices satisfying the algebra:

$$\alpha_i, \alpha_j = -\alpha_j\alpha_i \quad i \neq j, \quad \text{and} \qquad \alpha_i, \beta = -\beta\alpha_i,$$
$$(\alpha_i)^2 = \beta^2 = I_{4\times 4}$$

The matrices are given by:

$$\beta = \begin{bmatrix} I & 0 \\ 0 & -I \end{bmatrix}, \quad \alpha^i = \begin{bmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{bmatrix} \quad \text{for } i = 1, 2, 3$$

where I is the 2×2 identity matrix, and the Pauli matrices are:

$$\sigma^1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma^2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma^3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Taking for simplicity $\hbar = 1$ and $c = 1$ Now we replace the operators:

$$\hat{p}_i = -i\frac{\partial}{\partial x_i} \quad \text{and} \quad \hat{E} = i\frac{\partial}{\partial t}$$

leading to:

$$i\frac{\partial}{\partial t}\psi(t,\mathbf{x}) = \Big(-i\sum_{j=1}^{3}\alpha_j\frac{\partial}{\partial x_j} + \beta m\Big)\psi(t,\mathbf{x}) = 0.$$

$$\psi(t,\mathbf{x}) = \begin{bmatrix}\psi_1(t,\mathbf{x})\\ \psi_2(t,\mathbf{x})\\ \psi_3(t,\mathbf{x})\\ \psi_4(t,\mathbf{x})\end{bmatrix}$$

Edding the potential $V = V(x)$ we have leading to:

$$i\frac{\partial}{\partial t}\psi(t,\mathbf{x}) = \Big(-i\sum_{j=1}^{3}\alpha_j\frac{\partial}{\partial x_j} + \beta m + V(x)\Big)\psi(t,\mathbf{x}) = 0.$$

This equation incorporates both special relativity and quantum mechanics and predicts phenomena such as electron spin and antimatter.

5.1. **References: to know more on the Dirac equation and other fiend equations see:**

- Quantum Field Theory 2nd Edition by Lewis H. Ryder (Author)

5.2. **References: Mathematical approach to quantum mechanics Books.**

- Quantum Mechanics for Mathematicians (Graduate Studies in Mathematics Volume 95) UK ed. Edition by Leon A. Takhtajan (Author)
- Mathematical Methods in Quantum Mechanics: With Applications to Schrodinger Operators (Graduate Studies in Mathematics) (Graduate Studies in Mathematics, 157) 2nd Edition by Gerald Teschl (Author)

6. Spectral theory on the S-spectrum

There are many reasons to extend complex spectral theory to vector operators.

1. Quaternionic formulation of quantum mechanics. The interest in spectral theory for quaternionic operators is motivated by the paper:

- G. Birkhoff, J. von Neumann, *The logic of quantum mechanics*, Ann. of Math., **37** (1936), 823-843.
- S. Adler: *Quaternionic Quantum Mechanics and Quaternionic Quantum Fields.* Volume 88 of *International Series of Monographs on Physics.* Oxford University Press, New York (1995).

who showed that Schrödinger equation can be written basically in the complex or in the quaternionic setting.

2. Vector analysis. The gradient operator with nonconstant coefficients in n dimensions represents various physical laws, such as Fourier's law for heat propagation and Fick's law for mass transfer diffusion. This can be expressed as

$$T = \sum_{i=1}^{n} e_i a_i(x) \partial_{x_i}, \qquad x \in \Omega$$

the coefficients $a_1, \ldots, a_n$ belong to some function space. In particular for constant coefficients we have the well known gradient operators

$$T = \nabla = e_1 \partial_{x_1} + e_2 \partial_{x_2} + \ldots e_n \partial_{x_n}.$$

Here, $e_1, \ldots, e_n$ represent the imaginary units of the Clifford algebra $\mathbb{R}_n$ satisfying the relations

$$e_i e_j + e_j e_i = 0, \ i \neq j \qquad e_i^2 = -1.$$

3. Differential geometry. Let (M, g) be a Riemannian manifold of dimension n, and U a coordinate neighbourhood of M. In this neighbourhood, we can find an orthonormal frame of vector fields $E_1, \ldots, E_n$, such that the Dirac operator on (U, g) can be expressed as the $\mathcal{C}^\infty(U, \mathcal{H})$ differential operator

$$\mathcal{D} = \sum_{i=1}^{n} e_i \nabla^{\tau}_{E_i}, \tag{6.1}$$

where $\nabla^{\tau}_{E_i}$ denotes the covariant derivative with respect to the vector field E_i.

4. Hypercomplex analysis. The well-known Dirac operator and its conjugate

$$D = \partial_{x_0} + \sum_{i=1}^{n} e_i \partial_{x_i} \qquad \text{and} \qquad \overline{D} = \partial_{x_0} - \sum_{i=1}^{n} e_i \partial_{x_i},$$

are important operators.

6.1. Natural questions that arise in spectral theory for vector operators.

- The first question is: what is the notion of spectrum of a quaternionic operator or a Clifford operators? For example if we have the gradient

  $$\nabla = e_1 \partial_1 + \ldots + e_n \partial_{x_n}$$

  how to generalize the eigenvalue equation

  $$\nabla u = su? \qquad \nabla u = su?$$

  what is s? can it be $s = e_1 s_1 + \ldots + e_n s_n$???
- What are the resolvent operators, and what about the analyticity of the new resolvent operators?
- What is the analogue of the Riesz-Dunford functional calculus?
- What is the analogue of the Spectral Theorem ?

- What about the evolution operator in this setting?

The spectral theory on the S-spectrum has fundamental differences compared to classical complex spectral theory, because it is derived from the classical resolvent series expansion, without assuming that the operator and the spectral parameter commute. Specifically, for bounded operators $T : V \to V$ acting in a Clifford-Banach module

$$V = X \otimes \mathbb{R}_n$$

where for example

$$T = T_0 + T_1 e_1 + \cdots + T_n e_n \quad \text{for} \quad T_j : X \to X \quad \text{where X is a Real Banach space}$$

the series expansion of the S-resolvent becomes

$$\sum_{n=0}^{\infty} T^n s^{-n-1} = (T^2 - 2s_0 T + |s|^2)^{-1}(\overline{s} - T), \qquad |s| > \|T\|, \tag{6.2}$$

where

$$s = s_0 + s_1 e_1 + \cdots + s_n e_n \quad \text{is a paravector,}$$
$$|s| = \sqrt{s_0^2 + s_1^2 + \cdots + s_n^2} \quad \text{is its modulus,}$$

and

$$\overline{s} = s_0 - s_1 e_1 - \cdots - s_n e_n \quad \text{the conjugate of s.}$$

The reason why the value of the series (6.2) does not simplify to the classical resolvent operator $(s-T)^{-1}$ is due to the noncommutativity $sT \neq Ts$. The explicit value of the sum in equation (6.2) motivates that, even for unbounded, closed operators T, the spectrum must be associated with the invertibility of the operator

$$Q_s(T) := T^2 - 2s_0 T + |s|^2, \qquad \text{with } \operatorname{dom}(Q_s(T)) = \operatorname{dom}(T^2).$$

This leads us to the definitions of the *S-resolvent set* and the *S-spectrum*

$$\rho_S(T) := \left\{ s \in \mathbb{R}^{n+1} \;\middle|\; Q_s(T)^{-1} \in \mathcal{B}(V) \right\} \qquad \text{and} \qquad \sigma_S(T) := \mathbb{R}^{n+1} \setminus \rho_S(T),$$

where $\mathcal{B}(V)$ is the space of all bounded operators from V into itself.

Motivated by (6.2), we define for every $s \in \rho_S(T)$ the *left* and the *right S-resolvent operator*

$$S_L^{-1}(s,T) := Q_s(T)^{-1}\overline{s} - TQ_s(T)^{-1} \qquad \text{and} \qquad S_R^{-1}(s,T) := (\overline{s} - T)Q_s(T)^{-1}. \tag{6.3}$$

which is then be used in the definition of the S-functional calculus

$$f(T) := \frac{1}{2\pi} \int_{\partial U \cap \mathbb{C}_J} S_L^{-1}(s,T) ds_J f(s) \quad \text{and} \quad f(T) := \frac{1}{2\pi} \int_{\partial U \cap \mathbb{C}_J} f(s) ds_J S_R^{-1}(s,T), \tag{6.4}$$

for left resp. right slice holomorphic functions f. This is the Clifford analogue of the Riesz-Dunford functional calculus. Moreover, we remark that the spectral theorem has recently been extended from quaternionic operators and to fully Clifford operators.

Remark 6.1. *The S-spectrum was discovered only in 2006. From that year started the development of this new spectral theory.*

6.2. **Some references books on the spectral theory on the S-spectrum.**

(1) F. Colombo, I. Sabadini, D. C. Struppa, Noncommutative Functional Calculus. Theory and Applications of Slice Hyperholomorphic Functions, Progress in Mathematics, Vol. 289, Birkhäuser, 2011, VI, 222 p.
(2) F. Colombo, J. Gantner, D. P. Kimsey, Spectral Theory on the S-spectrum for quaternionic operators, Operator Theory: Advances and Applications, vol 270, 2018.
(3) F. Colombo, J. Gantner, Quaternionic closed operators, fractional powers and fractional diffusion process, Operator Theory: Advances and Applications, vol 274, 2019.

6.3. **Some recent papers on the spectral theory on the S-spectrum.**

(1) A. De Martino, S. Pinton, P. Schlosser, The harmonic H^∞-functional calculus based on the S-spectrum. J. Spectr. Theory 14, No. 1, 121-162 (2024).
(2) A. De Martino, S. Pinton, Properties of a polyanalytic functional calculus on the S-spectrum. Math. Nachr. 296, No. 11, 5190-5226 (2023).
(3) F. Colombo, A. De Martino, S. Pinton, I. Sabadini, Axially harmonic functions and the harmonic functional calculus on the S-spectrum. J. Geom. Anal. 33, No. 1, Paper No. 2, 54 p. (2023).